

Student No: .....



## ROSEVILLE COLLEGE

2007

### PRELIMINARY MATHEMATICS

### YEARLY EXAMINATION

**Time allowed:** Three hours (plus 5 minutes reading time)

**Directions to Candidates:**

- Attempt all questions
- All questions are of equal value
- All necessary working must be shown. Marks may not be awarded for answers unsupported by working
- Start each question on a new page
- Your rough work is to be attached to the back of this question paper

|       |                                         |  |  |
|-------|-----------------------------------------|--|--|
| 1     | Basic Arithmetic                        |  |  |
| 2     | Algebra and Surds                       |  |  |
| 3     | Equations                               |  |  |
| 4     | Geometry                                |  |  |
| 5     | Functions and Graphs                    |  |  |
| 6     | Trigonometry                            |  |  |
| 7     | Linear expression and the Straight Line |  |  |
| 8     | Introductory Calculus                   |  |  |
| 9     | The Quadratic Function                  |  |  |
| 10    | Locus and the Parabola                  |  |  |
| Total |                                         |  |  |

**Question 1 Basic Arithmetic Show all Working****12 marks**

- (a) Evaluate  $|-5| - |-8 - -5|$  (2)
- (b) Find , correct to 2 decimal places, the value of:  $\frac{14 \cdot 73 + 8 \cdot 96}{\sqrt{(5 \cdot 86)^2 - 2 \cdot 78}}$  (2)
- (c) Express  $0 \cdot 2\dot{3}$  as a fraction in lowest terms (2)
- (d) At a mobile phone sale offering 40% discounts, Julie paid \$135 for her phone. What was its original price? (2)
- (e) If  $a = -3$  and  $b = -2$  , find the exact value of  $a^3(a - b)$  (2)
- (f) Express  $3 \cdot 2^{25} \div 0 \cdot 014$  in scientific notation correct to 3 significant figures (2)

**Question 2 Algebra and Surds (Start a new page)****12 marks**

- (a) Simplify:  $2(a - 3) - 2(2a - 3)$  (2)
- (b) Simplify giving your answer in simplest surd form: (2)

$$\sqrt{8} + \sqrt{50} - 3\sqrt{18}$$

- (c) Express  $\frac{2\sqrt{3}}{5 - \sqrt{3}}$  with a rational denominator (2)

- (d) Simplify this algebraic fraction:

$$\frac{a+1}{3} - \frac{3-a}{2} \quad (2)$$

- (e) Fully factorise the expression  $3x^3 - 24$  (2)

- (f) Heron's formula for the area of a triangle with sides  $a$ ,  $b$  , and  $c$  is: (2)

$$A = \sqrt{S(S-a)(S-b)(S-c)} \quad \text{where } S = \frac{a+b+c}{2}$$

Find the area of a triangle with sides 5 cm, 6 cm and 7 cm.

(Give your answer in simplest surd form)

**Question 3 Equations****Start a new page****12 marks**

(a) Solve for x:  $4(x + 5) = -6(3 - x)$  (2)

(b) Solve this inequality and graph the solution on a number line: (3)

$$|2x - 3| \leq 5$$

(c) Solve the following simultaneous equations: (3)

$$2x + y = 12$$

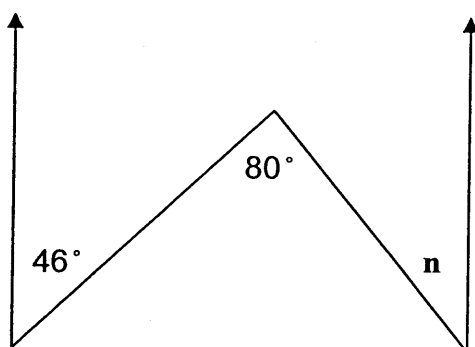
$$3x + 2y = 22$$

(d) Use the quadratic formula or otherwise to solve  $2x^2 - 3x - 1 = 0$ , (2)  
and give your answers correct to 1 decimal place.

(e) Solve the equation  $4^{3-x} = 8^x$  (2)

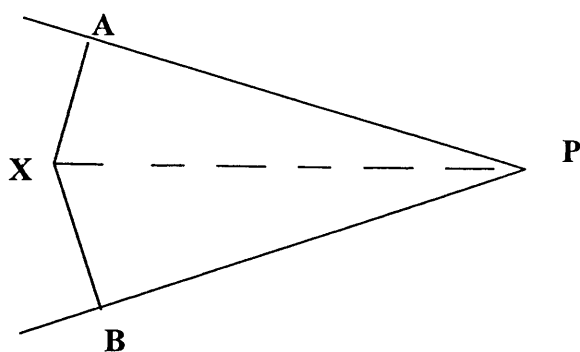
**Question 4 Geometry****Start a new page****12 marks**

(a) Find the value of n. You do not need to give reasons. (2)



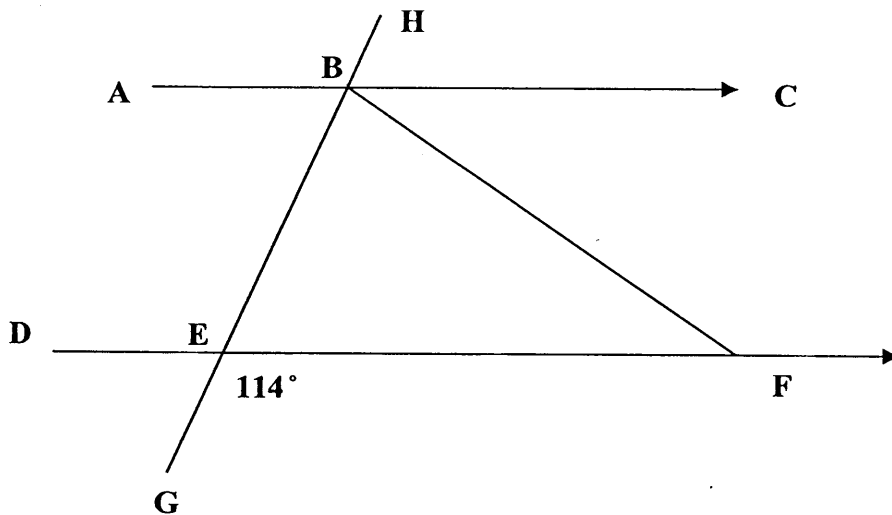
(b) In the diagram,  $AX = BX$ ,  $\angle PAX = \angle PBX = 90^\circ$ . Mark this on your diagram.

Using congruent triangles, prove that PX bisects  $\angle APB$  (3)



Question 4 (continued)

(c)



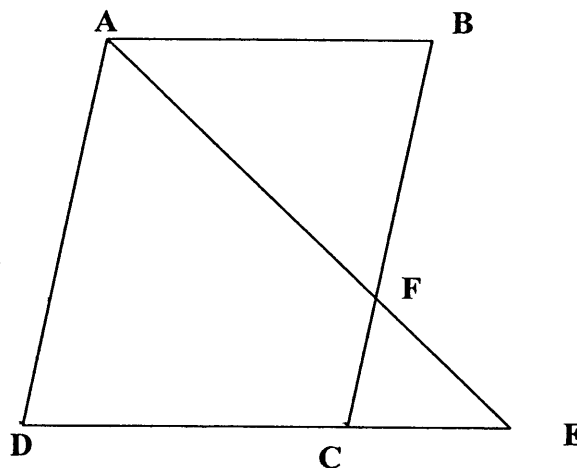
In the diagram,  $AC \parallel DF$ ,  $EF = BF$ ,  $\angle GEF = 114^\circ$ , and  $\angle CBF = x^\circ$ . (3)

- (i) Copy or trace the diagram, and mark the information on it.
- (ii) Find the value of  $x$ , giving complete reasons.

(d) ABCD is a parallelogram with DC produced to E, where  $AD = 120$  mm,  $CE = 40$  mm and  $BF = 70$  mm. (4)

(i) Show that  $\triangle ABF$  is similar to  $\triangle ECF$

(ii) Find the length of AB



**Question 5   Functions and Graphs   Start a new page****12 marks**

- (a) Draw clear sketches of each of the following showing the intercepts on the axes wherever appropriate. (8)

(i)  $x^2 + y^2 = 9$

(ii)  $2x + 3y = 6$

(iii)  $y = 2^x + 1$

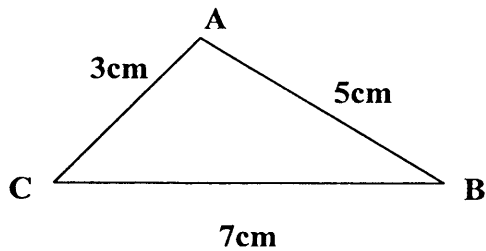
(iv)  $y = |x - 2|$

(b) Find  $\lim_{x \rightarrow 3} \frac{x^2 - 9}{x - 3}$  (2)

- (c) For the function  $y = 4 - x^2$ , find : (2)
- (i) its Domain;
  - (ii) its Range

**Question 6 Trigonometry****Start a new page****12 Marks**

- (a) Find the size of the largest angle in the triangle ABC (3)

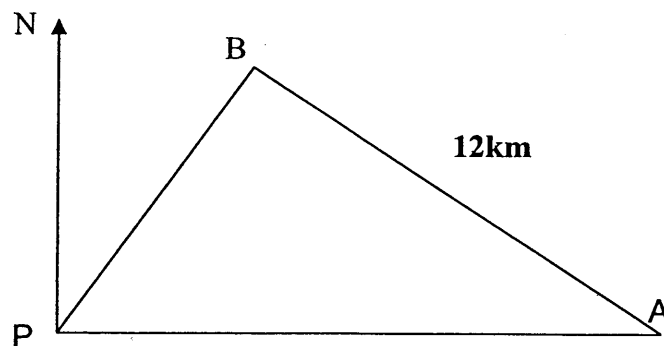


- (b) Find the exact value of : (3)

- (i)  $\sin 60^\circ$
- (ii)  $\cos 225^\circ$
- (iii)  $\cot(-30^\circ)$

- (c) On the sketch, two towns A and B are 12km apart on a straight road running northeast from A to B. A surveyor at P, due West of A, observes that the bearing of B is  $037^\circ$ .

- (i) Write down the size of  $\angle BAP$  and  $\angle APB$  (2)



- (ii) Calculate the distance of P from B, correct to 3 significant figures. (2)

- (d) Find all values of  $\theta$  such that  $\sqrt{3} \tan \theta = 1$  and  $0^\circ \leq \theta \leq 360^\circ$  (2)

**Question 7    Linear Expression and the straight Line    Start a new page    12 marks**

- (a) On a neat sketch, mark the origin O and A(5,0) B(8,4) and C(0,10) (1)  
Join A to B, B to C and C to A
- (b) Show that the line AB has equation  $4x - 3y - 20 = 0$  (2)
- (c) Show that AB is perpendicular to BC (2)
- (d) Show that  $AO = AB$  (2)
- (e) Find the area of the quadrilateral ABCO (2)
- (f) If D is the point (8,0), calculate the perpendicular distance of D from AB (2)
- (g) What is the equation of the circle with centre A (1)  
and which passes through both O and B?

**Question 8    Introduction to Calculus    Start a new page    12 marks**

- (a) Differentiate the following:
- (i)  $y = (3x - 2)^5$  (2)
- (ii)  $V = \frac{4}{3}\pi r^3$  (1)
- (b) Given that  $f(t) = \frac{2t-1}{3t+4}$ , find  $f'(-1)$  (2)
- (c) Find the equation of the tangent to the curve  $y = x^2 - 3x$  (3)  
at the point where  $x = 2$ .
- (d) Find the gradient of the normal to the curve to (2)  
the curve  $f(x) = 2x^3 - 4x - 5$  at  $(-1, -3)$
- (e) Find the size of the angle that the line  $8x - 5y = 9$  makes (2)  
with the positive x axis.

**Question 9    The Quadratic Function****Start a new page****12 marks**

- (a) The quadratic equation  $2x^2 - 5x + 8 = 0$  has roots  $\alpha$  and  $\beta$  (3)

Evaluate:

(i)  $\alpha + \beta$

(ii)  $\alpha\beta$

(iii)  $\alpha^3\beta^2 + \alpha^2\beta^3$

- (b) Without solving, show that the quadratic equation  $3x^2 + 2x - 5 = 0$  (1)  
has 2 different rational roots.

- (c) If the minimum value of the expression  $x^2 - 4x + k$  is 5, (2)  
find the value of  $k$

- (d) Find  $A$ ,  $B$  and  $C$  if  $6x^2 - 2x + 9 \equiv Ax(x-1) + B(x-1) + C$  (3)

- (e) For what values of  $k$  does the equation  $x + \frac{1}{x} = k$  have real roots? (3)

**Question 10 Locus and the parabola Start a new page****12 Marks**

- (a) What is the equation of the locus of a point P which moves so that  
Its distance from the origin is always less than or equal to 4 units (2)
- (b) By completing the squares, find the centre and radius of the circle  
whose equation is  $x^2 + y^2 - 2x + 6y - 15 = 0$  (3)
- (c) A parabola has its vertex at the point (3,1) and its focus at the point (3,3). (3)
- (i) What is its focal length?
- (ii) What is the equation of the directrix?
- (iii) What is the equation of the parabola?
- (d) A point P (x, y) moves so that the line PA is perpendicular to the line PB,  
where A = ( 1,5) and B = (-5,-3). Find the equation of this locus  
and describe it. (4)

SOLUTIONS + [MARKS] 2007 YR II Maths PRELIMINARY.

Q1 (a)  $5 - 3 = 2$  [2]

(b)  $4 \cdot 22$  [2]

(c)  $a = 0.2\dot{3}$

$10a = 2.3$

$100a = 23.\dot{3}$

$90a = 21$

$a = \frac{21}{90} = \frac{7}{30}$  [2]

(d)  $60\%$  of  $x = 135$

$20\%$  of  $x = 45$

$x = \$225$

[or unitary method] [2]

(e)  $(-3)^3(-3--2)$   
 $= -27 \times -1 = 27$  [2]

(f)  $3.04 \times 10^{14}$  [2]

Q2 (a)  $2a - 6 - 4a + 6 = -2a$  [2]

(b)  $\left. \begin{array}{l} \sqrt{8} = 2\sqrt{2} \\ \sqrt{50} = 5\sqrt{2} \\ 3\sqrt{18} = 3 \cdot 3\sqrt{2} \end{array} \right\} \begin{array}{l} 7\sqrt{2} - 9\sqrt{2} \\ = -2\sqrt{2} \end{array}$  [2]

(c)  $\frac{2\sqrt{3}(5+\sqrt{3})}{(5-\sqrt{3})(5+\sqrt{3})} = \frac{10\sqrt{3}+6}{25-3}$   
 $x = \frac{2(5\sqrt{3}+3)}{22}$  [2]  
 OR  
 $\rightarrow = \frac{(5\sqrt{3}+3)}{11}$

(d)  $\frac{a+1}{3} - \frac{3-a}{2} = \frac{2a+2-9+3a}{6}$   
 $= \frac{5a-7}{6}$  [2]

(e)  $3(x^3-8) = 3(x-2)(x^2+2x+4)$  [2]

(f)  $S = \frac{5+6+7}{2} = 9$

$A = \sqrt{9(4)(3)(2)} = \sqrt{36 \times 6} = 6\sqrt{6}$   
 [1] 6,  $\sqrt{36}$  [2]

Q3 (a)  $4x + 20 = -18 + 6x$

$38 = 2x$

$x = 19$

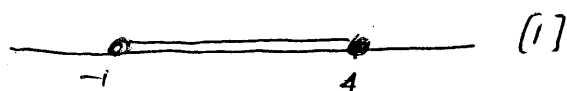
[2]

(b)  $|2x-3| \leq 5$

$2x-3 \leq 5$  or  $2x-3 \geq -5$

$2x \leq 8$   $2x \geq -2$

$x \leq 4$  [1]  $x \geq -1$  [1]



(c)  $2x + y = 12$  — (1)

$3x + 2y = 22$  — (2)

(1)  $\times 2$   $4x + 2y = 24$  — (3)

(3)  $-(2)$   $x = 2$   $\therefore x = 2$  [3]

sub  $y = 8$   $y = 8$

Check:  $6 + 16 = 22$

(d)  $2x^2 - 3x - 1 = 0$   $a=2$   
 $b=-3$   
 $c=-1$

$x = \frac{3 \pm \sqrt{9 - 4(2)(-1)}}{4}$

$= \frac{3 \pm \sqrt{17}}{4} = 1.8 \text{ or } -0.3$

[3]

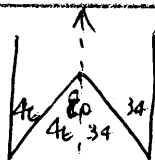
(e)  $4^{3-x} = 8^x$   
 $(2^2)^{3-x} = (2^3)^x$   
 $2^{6-2x} = 2^{3x}$

$6-2x = 3x$

$6 = 5x$

$\therefore x = \frac{6}{5}$  [2]

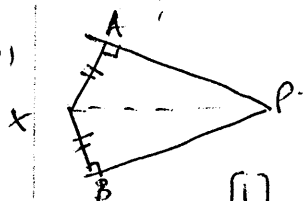
14. (a)



$$n = 34$$

[2]

(b)



In  $\Delta$ 's APX, BPX,

$$AX = BX \text{ (given)}$$

$$[1] \quad \hat{PAX} = \hat{PXB} = 90^\circ \text{ (given)}$$

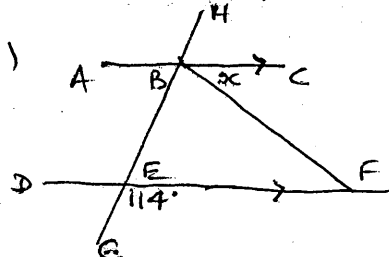
PX is common

$\therefore$  the  $\Delta$ 's are congruent (RHS) [1]

$$[1] \quad \therefore \hat{APX} = \hat{BPX} \text{ (cong angles in cong } \Delta\text{'s)}$$

$\therefore$  PX bisects  $\hat{APB}$ .

(c)



[1]

$$\hat{BEF} = 66^\circ \text{ (Suppl. adj } \angle\text{'s)}$$

$\Delta$  BFE is isosceles,  $\therefore$  base  $\angle$ 's equal

$$\therefore \hat{EFB} = 180 - 2 \times 66 = 48^\circ$$

[1] Then  $x = 48^\circ$  equal alternate  $\angle$ 's with  $AC \parallel DF$ .

(d) In  $\Delta$ 's ABF and CEF,

$$\hat{BAF} = \hat{CEF} \text{ (equal alt. } \angle\text{'s } AB \parallel DE)$$

$$\hat{ABF} = \hat{CEF} \text{ ( " " " )}$$

$$\therefore \hat{AFB} = \hat{CFE} \text{ (remaining } \angle\text{'s in } \Delta \text{ (angle sum theorem))}$$

$\therefore \Delta$ 's are similar (A.A.A.) [2]

$$\therefore \frac{AB}{40} = \frac{70}{50} = \frac{7}{5}$$

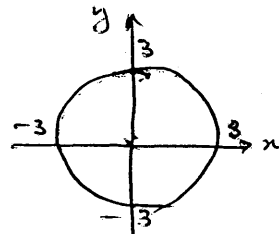
$$\therefore \hat{SAB} = 280$$

$$AB = 56 \text{ km.}$$

[2]

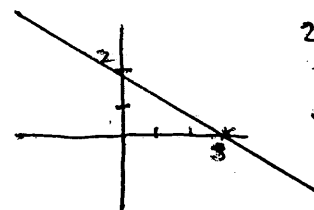
Q5

(a) (i)



[2]

(ii)



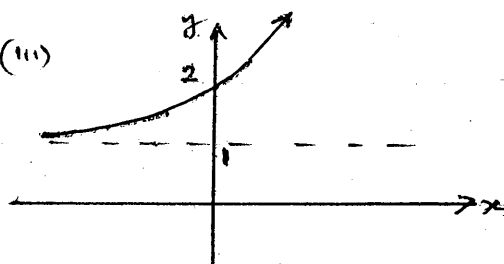
$$2x + 3y = 6$$

$$x = 0, y = 2$$

$$y = 0, x = 3$$

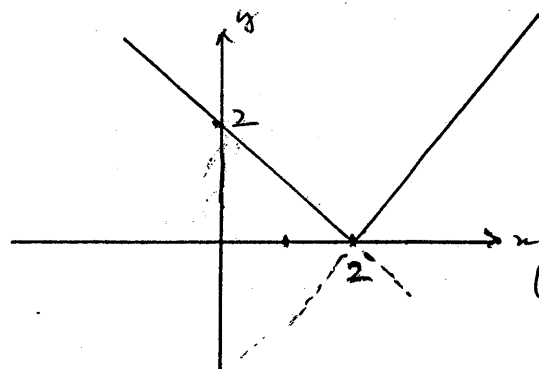
[2]

(iii)



[2]

(iv)



[2]

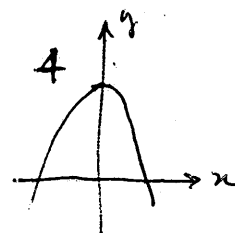
$$(b) \lim_{x \rightarrow 3} \frac{x^2 - 9}{x - 3} = \frac{0}{0}$$

$$= \lim_{x \rightarrow 3} \frac{(x-3)(x+3)}{(x-3)}$$

$$= \lim_{x \rightarrow 3} x + 3 = 6.$$

[2]

$$(c) y = 4 - x^2$$



$$D: x \in \mathbb{R} \quad [1]$$

$$R: y \leq 4 \quad [1]$$

Q6 (a)

$$\cos A = \frac{3^2 + 5^2 - 7^2}{2 \times 3 \times 5} = \frac{9 + 25 - 49}{30}$$

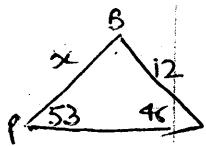
$$= \frac{-15}{30} = -\frac{1}{2}$$

$$\therefore A = 120^\circ \quad [3]$$

$$(b) (i) \sin 60 = \frac{\sqrt{3}}{2} \quad [1]$$

$$(ii) \cos 225 = \cos(180 + 45) \\ = -\cos 45 = \\ = -\frac{1}{\sqrt{2}} \quad [1]$$

$$(iii) \cot(-30) = \frac{1}{\tan -30} \\ = \frac{1}{-\frac{1}{\sqrt{3}}} \\ = -\sqrt{3} \quad [1]$$

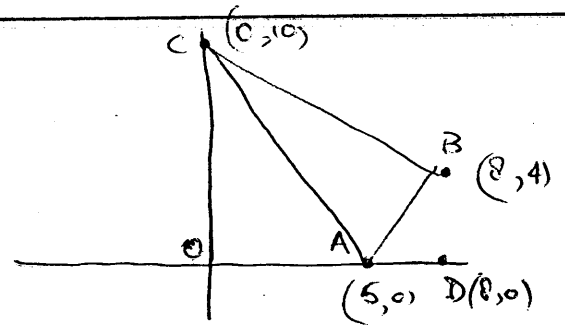


$$(c) (i) \hat{BAP} = 45^\circ \quad [1] \\ \hat{APB} = 53^\circ \quad [1]$$

$$(ii) \frac{PB}{\sin 45} = \frac{12}{\sin 53} \quad [1] \\ PB = \frac{12 \sin 45}{\sin 53} \quad [1] \\ = 10.6 \text{ km}$$

$$(d) \sqrt{3} \tan \theta = 1 \\ \therefore \tan \theta = \frac{1}{\sqrt{3}} \\ \therefore \theta = 30^\circ \text{ and } 180 + 30 \\ = 30^\circ \text{ and } 210^\circ \quad [2]$$

Q7 (a)



$$(b) \text{grad. of } AB = \frac{4}{3}$$

$$\text{eqn. is } y - 0 = \frac{4}{3}(x - 5)$$

$$3y = 4x - 20$$

$$4x - 3y - 20 = 0 \quad [2]$$

$$(c) \text{gradient of } BC = -\frac{6}{8} = -\frac{3}{4} \quad [2] \\ \text{" " } AB = \frac{4}{3}$$

$$\text{Since } -\frac{3}{4} \times \frac{4}{3} = -1, BC \perp AB.$$

$$(d) AO = 5 \text{ units } AB = \sqrt{(4-0)^2 + (8-5)^2} \\ = \sqrt{25} = 5$$

$$\therefore AO = AB. \quad [2]$$

$$(e) \text{Area of } \triangle OCA = \frac{1}{2} \times 5 \times 10 = 25 \\ \text{" " } \triangle ABC = 25 \text{ units}^2 \\ \therefore \text{Area of quadr. } OABC = 50 \text{ units}^2 \quad [2]$$

$$(f) d = \left| \frac{8 \times 4 - 3 \times 0 - 20}{\sqrt{4^2 + 3^2}} \right| \\ = \frac{32 - 20}{5} = 2\frac{2}{5} \text{ units} \quad [2]$$

$$(g) \text{Centre is } (5, 0) \text{ Radius is } 5 \\ \therefore \text{eqn. is } (x-5)^2 + (y-0)^2 = 5^2 \\ (x-5)^2 + y^2 = 25 \quad [1]$$

Q8 (a)(i)  $y = (3x-2)^5$   
 $y' = 5(3x-2)^4 \times 3$   
 $= 15(3x-2)^4$  [2]

(iii)  $V = \frac{4}{3}\pi r^3$   
 $\frac{dV}{dr} = \frac{12}{3}\pi r^2 = 4\pi r^2$  [1]

(b)  $f(t) = \frac{2t-1}{3t+4}$   
 $f'(t) = \frac{(3t+4) \cdot 2 - 3(2t-1)}{(3t+4)^2}$   
 $= \frac{6t+8-6t+3}{(3t+4)^2}$   
 $= \frac{11}{(3t+4)^2}$  [1]

So  $f'(-1) = \frac{11}{(-3+4)^2} = 11$  [1]

(c)  $y = x^2 - 3x$   
 $y' = 2x - 3$   
 if  $x = 2$ ,  $y = 4 - 6 = -2$  (2, -2)  
 if  $x = 2$ ,  $y' = 4 - 3 = 1$   
 $\therefore m = 1$ ,  $(x_1, y_1) = (2, -2)$

So eqn is  $y + 2 = 1(x - 2)$   
 $y + 2 = x - 2$   
 $y = x - 4$

(d)  $f(x) = 2x^3 - 4x - 5$   
 $f'(x) = 6x^2 - 4$   
 $f'(-1) = 6(-1)^2 - 4 = 2$  [1]  
 $\therefore$  Normal has gradient  $-\frac{1}{2}$  [1]

(e)  $8x - 5y = 9 \Rightarrow y = \frac{8}{5}x - \frac{9}{5}$   
 $\tan \theta = \frac{8}{5}$ ,  $\therefore \theta = 58^\circ$

Q9  $2x^2 - 5x + 8 = 0$   
 (a)(i)  $\alpha + \beta = \frac{-b}{a} = \frac{5}{2}$  [1]  
 (ii)  $\alpha\beta = \frac{c}{a} = \frac{8}{2} = 4$  [1]  
 (iii)  $\alpha^2\beta^2(\alpha + \beta) = 4^2 \times \frac{5}{2}$   
 $= 40$  [1]

(b)  $3x^2 + 2x - 5 = 0$

$\Delta = 4 - 4(3)(-5) = 4 + 60 = 64$

Since 64 is a perfect square, roots are RATIONAL and DIFFERENT [1]

(c)  $x^2 - 4x + K$   
 Min. value is when  $x = \frac{-b}{2a} = \frac{4}{2} = 2$   
 if  $x = 2$ ,  $2^2 - 4 \times 2 + K = 5$   
 $4 - 8 + K = 5$ ,  
 $\therefore K = 9$  [2]

(d)  $Ax(x-1) + B(x-1) + C = 6x^2 - 6x + 1$   
 Equating coefficients:  $A = 6$  [1]  
 if  $x = 1$ ,  $C = 6 - 2 + 9 = 13$   
 if  $x = 0$ ,  $12(1) + B(1) + 13 = 24 - 4 + 9$   
 $B + 25 = 29$   
 $B = 4$  [1]

$\therefore A = 6, B = 4, C = 13$

(e)  $x + \frac{1}{x} = K \rightarrow x^2 + 1 = Kx$   
 $x^2 - Kx + 1 = 0$  [1]

For real roots,  $\Delta \geq 0$

$K^2 - 4(1)(1) \geq 0$   
 $K^2 - 4 \geq 0$  [1]

$(K-2)(K+2) \geq 0$

$K \leq -2$  and  $K \geq 2$  [1]



Q10

(a)

$$x^2 + y^2 \leq 4^2$$

$$\text{u } x^2 + y^2 \leq 16$$

[2]

[1] for  $= 16$ , [1] for  $<$ 

$$(b) \quad x^2 + y^2 - 2x + 6y - 15 = 0$$

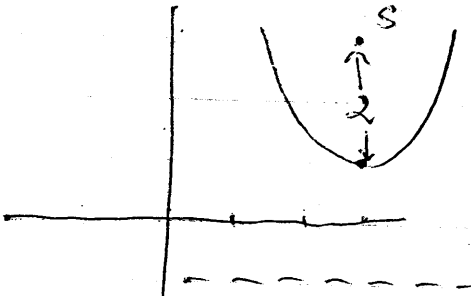
$$x^2 - 2x + 1 + y^2 + 6y + 9 = 15 + 1 + 9$$

$$(x-1)^2 + (y+3)^2 = 25$$

Centre is  $(1, -3)$  radius = 5.

[3]

(c)



$$(i) \quad a = 2 \quad [1]$$

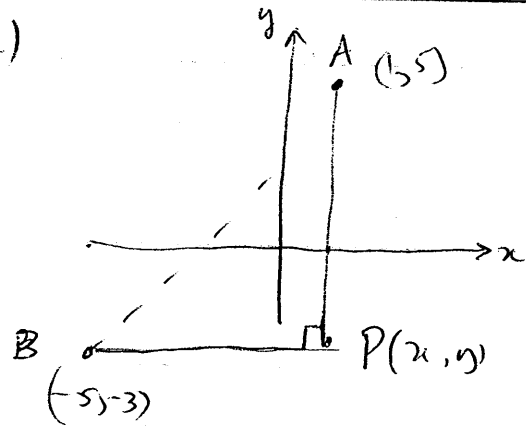
$$(ii) \quad \text{direction is } y = -1 \quad [1]$$

(iii) Parabola is

$$(x-3)^2 = 4 \times 2(y-1)$$

$$\text{u } (x-3)^2 = 8(y-1) \quad [2]$$

(d)



$$PA \perp PB$$

$$\text{gradient of } PA = \frac{y-5}{x-1}$$

$$\text{gradient of } PB = \frac{y+3}{x+5}$$

$$\text{So } \frac{y-5}{x-1} \times \frac{y+3}{x+5} = -1$$

[4]

$$y^2 - 2y - 15 = -x^2 - 4x + 5$$

$$x^2 + y^2 + 4x - 2y - 20 = 0$$

$$x^2 + 4x + 4 + y^2 - 2y + 1 = 20 + 4 + 1$$

$$(x+2)^2 + (y-1)^2 = 25$$

Circle, centre  $(-2, 1)$  radius 5.